

Coding open-ended responses with large language models: Evidence from three economic experiments*

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Abstract

This paper examines whether large language models (LLMs) can reliably code open-ended responses and free-form communication data in experimental economics. We apply three LLMs (GPT, Claude, and Gemini) under alternative prompt designs and temperature settings to data from three published experiments that differ in coding structure, the degree of textual ambiguity, and the role of communication in the experimental design. In a trust game where the coding task targets explicit commitment language, all three models achieve strong agreement with human benchmarks (Cohen's $\kappa > 0.80$) and replicate the finding that promise-making predicts cooperative behavior. In thematic coding of explanations for discrepancies between self- and peer-reported AI use, LLMs reliably recover the dominant category but over-identify rare inductively derived categories by a factor of three to five. In multi-label coding of chat communication in a coordination experiment, agreement is weaker and downstream regressions do not replicate consistently, with some coefficients reversing sign. Model choice and temperature are not first-order determinants of performance. The primary constraint is the structural fit between the coding scheme and LLM capabilities: LLMs succeed when categories are deductively defined and explicitly marked in the text, and fail when coding relies on pragmatic inference or inductively derived themes. These findings have direct implications for measurement reliability and analytical transparency in experimental research that uses coded text.

JEL Codes: C90, C92, D88.

Keywords: Large language models, Text coding, Reproducibility, Open-ended responses, Measurement reliability.

*This research was funded by Pontificia Universidad Javeriana (Project 10844). We thank Jordi Brandts, David Cooper, Alex Imas, and Yier Ling for generously sharing their data, and Ali Ozkes for kindly providing data from his study [Hanaki and Ozkes \(2023\)](#). *Data availability*: our analysis code is available on [GitHub](#), and the replication package is available at OSF (<https://osf.io/rvf6e>).

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1 Introduction

Textual data occupy an intermediate position between observed choices and latent mental states: they are observable behavior, yet they often encode reasoning, beliefs, and motivations that choices alone leave ambiguous. Even in experiments without direct participant communication, researchers often collect open-ended responses to understand how subjects interpret the experimental environment, justify their choices, or form beliefs about other participants. In many settings, these responses contain information central to the underlying behavioral question: motives, persuasion attempts, norm perceptions, strategic misunderstandings, and reactions to treatment conditions are often only partially observable in choices. However, because such data are rich and unstructured, they are costly to analyze systematically. Transforming free-form text into structured analyzable variables typically requires teams of research assistants to read, interpret, and manually code responses, thereby transforming the responses into analyzable quantitative measures. This makes text-intensive projects difficult and costly to scale, costly to replicate, and difficult to standardize. As [Gentzkow et al. \(2019\)](#) note, open-ended responses are theoretically informative but challenging to code systematically because of their heterogeneity and complexity. This challenge also raises concerns about reliability: manual coding introduces analytical flexibility through coder discretion, and the absence of standardized and replicable coding protocols makes it difficult to verify, replicate, or extend empirical findings that depend on coded text.

This paper examines whether large language models (LLMs) can code experimental text with sufficient reliability to serve as substitutes for, or complements to, human coders. We evaluate whether LLM-generated codes replicate benchmark human-coded classifications and support the same empirical conclusions regarding treatment effects and behavioral correlates. Our focus is not on whether LLMs can summarize or interpret text in a general sense, but whether they generate research-quality coded measures from participants' open-ended responses. A positive answer would imply that LLM-based coding can improve the transparency and reproducibility of text analysis in experimental economics by replacing discretionary manual coding procedures with a documented, replicable protocol.

LLMs are increasingly viewed as a scalable approach to analyzing language-rich data and reducing the cost of processing unstructured information ([Charness et al., 2023](#); [Ludwig et al., 2025](#)). [Charness et al. \(2023\)](#) characterize this as a turning point in which AI tools may reshape multiple stages of the research process, from design to implementation to analysis. Beyond their use in coding and analysis, LLMs are also being studied as experimental counterparts in their own right; a recent example examines trust behavior toward ChatGPT as a trustee in a trust game ([de Boer et al., 2026](#)). Yet the existing discussion

remains largely conceptual, and evidence on LLM-based coding in experimental settings is limited. Experimental text provides a demanding test case because coding often depends on subtle distinctions among strategic content, promises, persuasive language, and inductively derived themes. Whether LLMs can recover these distinctions with sufficient reliability to support empirical inference remains an open question. This is also a question of measurement consistency: if LLM codes and human codes lead to different conclusions, then the empirical findings of a study become contingent on the coding method, which itself constitutes a form of analytical flexibility that existing research standards do not explicitly address.

We address this question using three empirical applications drawn from previously published experimental studies in which human coding plays a central role. The first is [Brandts and Cooper \(2025\)](#), which examines chat communication in a coordination game with asymmetric information. The second is [Ederer and Schneider \(2022\)](#), which analyzes free-form promises in a repeated trust game, a setting in which messages appear strategically simple but may encode psychologically rich commitments. The third is [Ling et al. \(2025\)](#), which examines students' explanations for discrepancies between self-reported and peer-reported AI use and employs thematic coding of open-ended responses. Taken together, these applications allow us to compare LLM and human coding across a heterogeneous set of tasks that differ in the structure of the coding scheme, the degree of textual ambiguity, and the substantive role of text data within the experimental design.

This paper makes three main contributions. First, it provides systematic evidence on the extent to which LLMs replicate benchmark human-coded classifications in experimental data, using human labels as the gold standard. Second, it shows that LLM performance is not uniform across coding tasks: agreement depends on whether categories are tightly defined, binary versus multi-label, and explicitly marked in the text versus interpretive. Third, it moves beyond agreement metrics to evaluate whether LLM-generated codes support the same substantive conclusions as the original papers. This distinction is important for evaluating reliability: imperfect agreement need not imply unusable measurement, but high mechanical agreement need not guarantee that substantively meaningful inferences are preserved. Demonstrating this distinction directly addresses concerns about the robustness of empirical findings to analytic choices in text coding.

Our findings highlight both the promise and the limits of LLM-based coding. In some settings, LLMs closely reproduce human judgments and appear capable of generating usable research measures at substantially lower cost. In other settings, especially where coding relies on inductively developed themes or nuanced interpretation, agreement deteriorates and the choice of coding protocol becomes consequential. The broader implication is that LLMs are best understood not as automatic replacements for human interpretation, but as

tools whose reliability depends on the structure of the underlying measurement problem.

The remainder of the paper proceeds as follows. Section 2 situates the paper within the literature on experimental communication, human coding, and LLM-assisted text analysis. Section 3 presents the LLM coding framework, including model selection, prompt design, and implementation under alternative prompting strategies and temperature settings. Section 4 defines the validation framework and the metrics used to compare LLM-generated and human-generated coding. Section 5 introduces the three empirical applications, reports agreement between LLM-generated and human-generated coding for each application, and examines whether LLM-generated labels preserve the substantive conclusions of the original studies. The final section discusses what predicts LLM coding performance across tasks, provides practical guidance for researchers, and discusses broader implications for the use of LLM-based coding in experimental economics.

2 Related literature

Communication and open-ended responses have long been central to experimental economics, enabling researchers to test predictions from influential theories of human behavior and to develop a richer picture of the variables individuals consider relevant when making choices or forming beliefs (Haaland et al., 2025). Open-ended measurement captures key features of real-world communication and allows individuals to express their considerations, feelings, and experiences in their own words, without being constrained by predefined response options (Haaland et al., 2025). An early and ongoing experimental literature provides systematic evidence on the importance of natural language communication for economic outcomes, demonstrating that allowing communication can foster cooperation or collusion in a variety of contexts (Houser and Xiao, 2007; Andreoni and Rao, 2011; Brandts and Cooper, 2025; Baranski and Haas, 2023; Zultan, 2012). Researchers use such qualitative data to measure motives and reasoning underlying decisions, as well as mental models, narratives, and attention allocation, decision-relevant factors that are often unobserved in traditional datasets (Haaland et al., 2025; Chopra and Haaland, 2023). Chat logs from experiments can yield insights into participant behavior, uncertainty, and cognitive processes, providing an under-the-hood perspective that can generate novel hypotheses (Ling et al., 2025). A full understanding of human behavior requires not only quantitative data, typically collected from observed choices, but also qualitative data on how people perceive and reason about the world (Chopra and Haaland, 2023).

Despite their richness, experimental text data present substantial challenges for systematic coding at scale, limiting their use even when such data are collected. Open-ended

responses provide detailed and nuanced information and often uncover unexpected themes, but their unstructured nature also creates greater scope for both classical and nonclassical measurement error. In particular, researchers may worry that analyses based on such data will be viewed as questionable because human coding may be influenced by biases that align with researcher hypotheses. Additionally, some respondents may be unwilling to exert effort, and even when they do, their responses may remain ambiguous and difficult to interpret (Haaland et al., 2025). While open-ended social simulations can be highly informative, selecting and analyzing their outcomes can be difficult, as researchers may need to sift through thousands of lines of unstructured text to identify meaningful patterns (Charness et al., 2023). Previous research documents a progression from manual coding (Andreoni and Rao, 2011; Burchardi and Penczynski, 2014; Nielsen et al., 2019; Ederer and Schneider, 2022; Brandts and Cooper, 2025) to semi-automated and fully automated approaches using, for instance, machine learning (Penczynski, 2019; Hausladen et al., 2024; Chen et al., 2026), and, more recently, to the use of LLMs as classifiers (Korinek, 2023; Chopra and Haaland, 2023; Celebi and Penczynski, 2024; Bermejo et al., 2025; Haaland et al., 2025). In this context, LLMs offer the potential to standardize text analysis at scale, reduce concerns about subjectivity and coder reliability, and provide a more cost-effective alternative that may reduce disparities between researchers with substantial funding and those with fewer resources, including junior scholars and researchers in developing countries.

However, developing a coding scheme is costly and requires researchers to make subjective choices that may not be fully replicable and may also be prone to bias (Haaland et al., 2025). Inductive coding begins with the data and develops codes based on themes that emerge directly from open-ended responses, whereas deductive coding uses existing knowledge and theory to define a coding scheme (Haaland et al., 2025). Best practices call for multiple researchers to independently review responses to identify appropriate codes and then to work together to develop a final set of codes that remains close to respondents' own language. A qualitative codebook that includes the final set of codes along with positive and negative examples illustrates how the codes should be applied, and all coders should review the codebook carefully, participate in joint training, and then independently hand-code the same data (Chopra and Haaland, 2023). Research assistants typically receive extensive training before beginning the task, including multiple rounds of double- and triple-coding, followed by group discussions about how to reconcile coding differences (Chopra and Haaland, 2023). Ensuring that coders are unaware of the research hypothesis can reduce the potential for bias, and double-coding responses with discrepancies resolved through a third coder can reduce measurement error (Haaland et al., 2025). Inter-coder reliability measures the extent of agreement across coders, with common measures including Cohen's kappa and Krippendorff's alpha. As noted by Haaland et al. (2025), Miles and Huberman (1994)

suggest 80% agreement on 95% of codes, whereas Neuendorf (2017) proposes thresholds of 90% as widely acceptable and 80% as acceptable in many applications.

Recent evidence suggests that large language models can code open-ended data reliably and reproducibly, outperforming crowd workers and making them a low-cost and easily scalable alternative to human coding (Bermejo et al., 2025; Haaland et al., 2025; Celebi and Penczynski, 2024). LLMs offer several attractive performance features: they generate immediate outputs, provide detailed justifications on demand, and produce consistent classifications that are not subject to fatigue (Celebi and Penczynski, 2024). In experimental economics, Celebi and Penczynski (2024) conducted a similar exercise to ours and found that GPT-4 outperforms GPT-3.5 and achieves high levels of agreement with human coders, reaching nearly 90% accuracy in applications involving promises and up to 73% in more complex tasks such as identifying strategic thinking, which requires understanding and applying specific game-theoretic concepts and interpreting implicit linguistic patterns. Human-validated AI coding can serve as a cost-effective substitute for labeling interview transcripts, with GPT-4o achieving global interrater agreement of 86% between AI and human coders, matching human-human agreement (Chopra and Haaland, 2023). LLMs consistently outperform outsourced human coders, particularly in tasks requiring deep contextual understanding, and offer a cost-effective alternative for sophisticated text analysis through simple API-based workflows and modest operational costs (Bermejo et al., 2025). Traditional methods such as BERT-based classification (Seiler, 2025) require significant training data, and their performance deteriorates for more nuanced concepts, whereas LLMs can reduce the need for extensive labeled training data and train-test separation, making them advantageous in low-resource environments (Bermejo et al., 2025). However, although frontier LLMs can provide a low-cost and easily scalable alternative to human coding, their performance can depend on the quality of the prompting, the type of LLM used, and the complexity of the setting, and it remains essential to validate AI-generated coding against human benchmarks to guard against hallucinations and systematic model bias.

3 The LLM coding framework

To systematically evaluate the performance of LLMs in coding open-ended responses, we developed a modular framework designed for reproducibility and scalability. This framework can be visualized in Figure 1, which describes the six steps ranging from the submission of the input file to the download of the coding results.

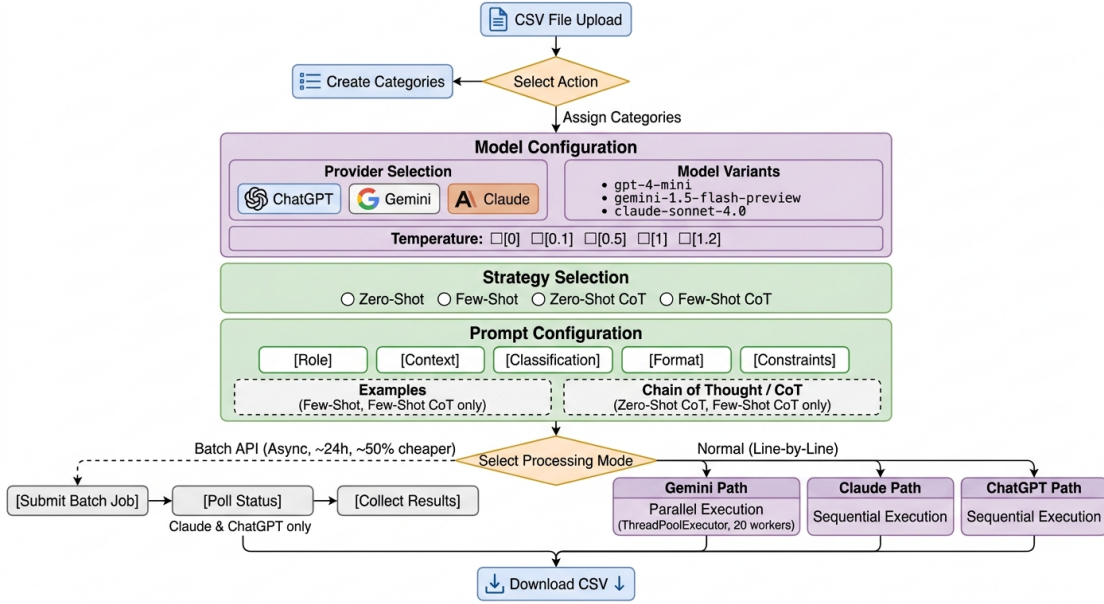


Figure 1. Flow diagram of the LLM coding procedure

The first step is to upload a CSV input file. This file must contain, at a minimum, the individual textual inputs to be analyzed, with the relevant input column explicitly specified. After the file is uploaded, one can choose between *assigning categories* to each open-ended response or *creating categories* based on the whole set of responses. If the last option is chosen (*Create Categories*), the result will be a CSV file with the categories created by the LLM and a definition for each label. On the other hand, *Assign Categories* will generate a CSV file with all the responses classified, and additional columns specified by the user.

This section only describes how to assign categories using the developed framework, as the present paper focuses on evaluating LLM performance in classifying responses from the three empirical applications presented herein. Nonetheless, in other contexts, the *Create Categories* option may be useful for initially defining the labels that will later be employed for classification. Crucially, this option follows a structured procedure that makes the category-creation process explicit, documented, and replicable: the researcher specifies the coding objective, provides illustrative examples of the target construct, and the framework derives candidate categories through a systematic inference process rather than relying solely on researcher intuition. This reduces the risk that the resulting categories are overly ad hoc or idiosyncratic, since any researcher following the same inputs and steps can reproduce the same label set. Additionally, because *Create Categories* follows a procedure similar to that of the *Assign Categories* option, the explanation that follows will also facilitate understanding of this functionality.

For *Assigning Categories*, the second stage of the framework involves selecting a specific

LLM and its corresponding version. For this research, we selected models from the Gemini, GPT, and Claude families, given their prominence and widespread use in the current academic landscape. By incorporating multiple models, we aim to provide a robust comparison of performance across diverse textual inputs and experimental contexts.

Next, we proceed to the Prompt Configuration module. This third step constructs the prompt for the model using a template composed of five sections. The first section is the *Role Definition*, which establishes the LLM’s identity as an expert research assistant specializing in the qualitative analysis of behavioral economics experiments. Following the *Role Definition*, the *Context Setting* provides an overview of the experimental design. Specifically, we describe the game structure, treatment conditions, communication type, control mechanism, and other relevant details required for the model to understand the underlying experimental logic without disclosing the results. Next, the *Classification Instructions* explicitly define the categories used for the coding task, together with the language and structure of the participants’ responses. To ensure that the output is suitable for automated analysis, the *Format Specification* defines the required response format for each classified response. In this case, we required the output to follow a JSON structure. The fifth and final section is *Constraints*. Here, we specify how each category must be classified; for example, whether the classification is binary or allows additional options. We also describe any conditional dependencies among categories. One example is the following: if ‘A’ is false, then ‘B,’ ‘C,’ and ‘D’ must also be false.

The *Classification Strategy* forms part of the prompt structure we developed. Depending on the type of classification, additional specifications may be incorporated. In this case, the application allows users to choose among four strategies: zero-shot, few-shot, zero-shot Chain-of-Thought (CoT), and few-shot CoT. For zero-shot classification, for example, the basic prompt structure described above is sufficient, whereas few-shot classification requires providing the LLM with one or more examples of previously classified responses (Brown et al., 2020). Zero-shot CoT and few-shot CoT operate similarly to the previous two approaches but additionally instruct the model to generate intermediate reasoning steps or explain the logic underlying the final classification (Wei et al., 2022).

This prompt structure was based on the template proposed by Penczynski (2019). They combine role specification, contextual information, classification instructions, examples, output constraints, and other modular components into a unified prompting framework. The purpose of generating a fixed structure is to reduce prompt-design confounds, ensure consistency across tasks, and better isolate differences in performance attributable to task complexity rather than prompt variation. It is also important to take into account that the prompt structure is not the only aspect that may influence how optimal the results are; also the word choice may play a role in the LLM’s performance Penczynski (2019).

Once the prompt template has been specified, the next step is to select the temperature setting for the LLM. Temperature is an important parameter because it controls the randomness of the model’s output: higher temperatures increase variability in responses, whereas lower temperatures produce more deterministic and reproducible classifications. Although the available temperature range may vary across models, values closer to zero generally yield more stable outputs. Evaluating different temperature settings allows us to examine the extent to which this parameter influences classification results and to assess how closely the model-generated classifications at each temperature align with the original human-coded annotations.

The final stage is the *Processing Modes*. In this module, candidate responses are processed to obtain their corresponding classifications. All Gemini, ChatGPT, and Claude models can assign classifications on a per-response (or per-observation or line-by-line) basis, while ChatGPT and Claude also allow processing in batch mode, in which multiple observations are submitted and classified within a single asynchronous job. This mode is particularly useful for large-scale annotation tasks, as it can reduce computational costs, API overhead, and processing time (OpenAI, 2024). In the case of per-observation processing, the classification results can be downloaded as a CSV file. Additionally, if the classification process is interrupted, whether due to connection issues or because the researcher chooses to stop execution, the framework saves progress in a helper file, allowing the process to resume from the last completed observation.

This framework was developed with the aim of facilitating the creation of categories and the classification of open-ended responses for researchers who need to carry out this process on a given dataset, free of charge. The framework was implemented as a web-based application to ensure that the procedure remains consistent and that results can be obtained in a user-friendly manner. The code is available in this [GitHub repository](#), which also contains a README file explaining how to run and use the tool. The Online Appendix provides screenshots of the online application.

4 Validation approach and metrics

We computed a set of performance and agreement metrics, including accuracy, precision, recall, F1-score, Cohen’s kappa (Cohen, 1960), and Krippendorff’s alpha (Krippendorff, 2011), to assess both how closely model-generated classifications align with the original human-coded labels and the reliability of the models’ classifications. The first four metrics evaluate how well the LLMs classify responses relative to the true classifications: accuracy measures the proportion of correct classifications, precision the proportion of positive classifications

that are actually correct, recall the proportion of actual positive cases that are correctly identified, and the F1-score the harmonic mean of precision and recall. Because accuracy can obscure poor performance in minority classes when categories are imbalanced, we report precision and recall as macro averages over both classes. Formal definitions and illustrative examples are provided in the Online Appendix.

Besides these metrics, Cohen’s kappa and Krippendorff’s alpha are commonly used in the literature on the classification of open-ended responses in behavioral experiments. Both are measures of inter-rater reliability, meaning that they assess the extent of agreement between coders beyond what would be expected by chance; Cohen’s kappa applies to two raters, while Krippendorff’s alpha accommodates more than two. In this study, the selected LLM is treated as one coder and the ground-truth classification as the other. Krippendorff’s alpha is used in two ways: in the application where classifications are generated by three human coders, the LLM is included as an additional (fourth) coder, and in the internal consistency analyses, each temperature run of a given model is treated as an independent rater.

5 Three empirical applications

5.1 Application 1: Free-form chat in managerial coordination game (Brandts and Cooper, 2025)

In Brandts and Cooper (2025), the authors study how effective different types of communication and managerial control are in facilitating coordination within organizations. They consider a setting characterized by information asymmetries and preference conflicts between managers and agents. To analyze this environment, they employ a game-theoretic framework they term the Manager-Agent (MA) game. The MA game is a three-player coordination game involving one manager (M) and two agents (A1 and A2). In the game, all participants, regardless of role, benefit from coordinating on a common action, but they have differing preferences over which action should be chosen. The state of the world, which determines the efficient outcome, varies; agents observe this state, whereas the manager does not. Given preference heterogeneity and the manager’s informational disadvantage, agents have an incentive to misreport the state of the world to influence the manager’s decision toward their preferred outcome. Central to the experimental design, the authors vary two factors to identify their causal effects: (1) whether agents’ actions are determined by managerial control (the manager imposes decisions) or by delegation (agents choose their own actions); and (2) the type of communication available, including no communication, structured communication (i.e., pre-specified message spaces), or free-form chat.

The treatments of their experiment are: No Communication with Delegation (NC-D); Structured Communication between Agents with Delegation (SC/S-D); Structured Communication with Advice and Delegation (SC/A-D); Structured Communication with Managerial Control (SC-MC); Chat between Agents with Delegation (CH/S-D); Chat with Advice and Delegation (CH/A-D); and Chat with Managerial Control (CH-MC). An additional treatment, Extended Communication and Managerial Control (EC-MC), was later introduced to isolate the marginal effect of free-form communication.

In this paper, we focus on treatments involving free-form chat. In the original paper, [Brandts and Cooper \(2025\)](#) develop a systematic coding scheme to analyze the content of free-form chat messages in the Manager-Agent (MA) game. They employ an iterative review of a random sample of conversations to identify relevant patterns and categories. Two research assistants independently code all chat conversations using a binary coding scheme in which each message line is assigned a value of 1 if it contains content relevant to a given category and 0 otherwise, with no restriction on the number of categories assigned to each message. The final dataset contains 965 conversations with 15 labels coded at conversation level. For categories exhibiting low inter-rater agreement (Cohen’s kappa < 0.5), a third, extensively trained coder recoded those categories to enhance reliability. Coders are explicitly instructed to code content objectively, without considering potential effects, and remain blind to the authors’ hypotheses. The coding scheme encompasses multiple categories, including suggestions (safe or efficient outcomes), agreement, discussion topics (coordination, fairness, efficiency, experimental rules, and gameplay), information revelation (truthful disclosure, deception, and contradictions), and the frequency of managerial communication.

5.1.1 [Brandts and Cooper \(2025\)](#): validation metrics

Figure 2 presents the inter-rater reliability, measured by Krippendorff’s α , for each category. Some categories exhibit high agreement among the three human coders. However, other categories appear more difficult to code and yield lower agreement scores. These α values help distinguish cases in which low model performance reflects LLM misclassification from cases in which the category itself is difficult to code, as indicated by low agreement among human coders.

Before evaluating LLM performance, we partition the 15 categories according to their prevalence in the evaluation sample. Two categories, *falsehood* and *contradict*, contain no positive cases, rendering agreement statistics undefined; accordingly, we exclude them from all analyses. The absence of *falsehood* and *contradict* in our sample reflects the structure of the treatments we evaluate, which focus on free-form chat with delegation and managerial

control, and these categories may be more prevalent in other treatment arms. One additional category, *discuss_howtoplay*, contains only nine positive cases, below the threshold of 10 required for reliable estimation. We report results for this category in the Online Appendix but exclude it from the main analysis. This leaves 12 evaluable categories.

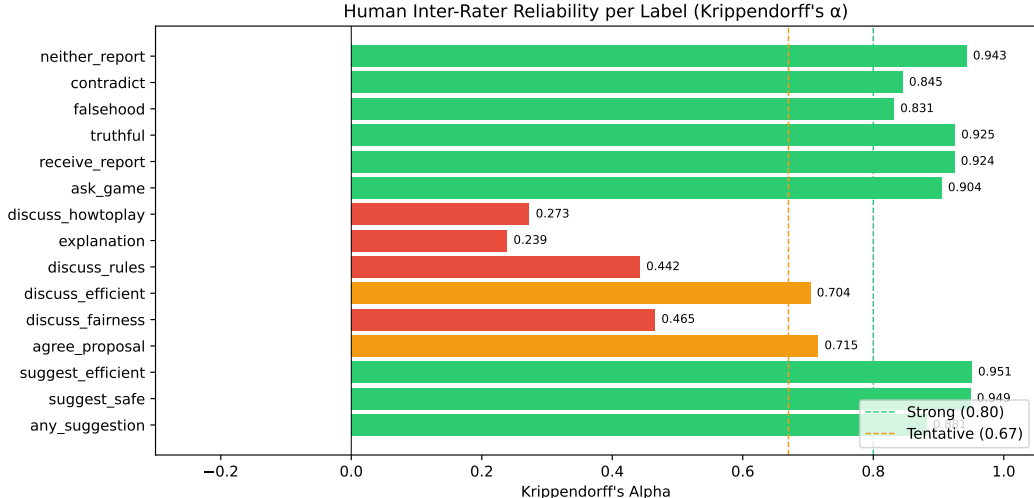


Figure 2. Human Inter-Rater reliability per category (Krippendorff’s α)

Figure 2 shows that the evaluable categories fall into two broad groups. Eight are *behaviorally concrete*: they capture whether agents make or agree to specific proposals, ask about the game state, or truthfully report private information, and human coders exhibit high agreement on these categories ($\alpha \geq 0.70$). The remaining four are *interpretively ambiguous*: they concern broader conversational themes, such as fairness, efficiency, or rules, or require inference about communicative intent (*explanation*), resulting in substantially lower human agreement (α ranging from 0.24 to 0.47). This distinction is important because LLMs cannot reasonably be expected to systematically exceed human inter-rater reliability; the category-specific human α therefore provides the relevant benchmark ceiling, rather than a uniform threshold such as 0.80.

Then, for each evaluable tag, we select the best-performing configuration for each model across all combinations of prompting strategy (zero-shot vs. few-shot) and temperature, and report the resulting Krippendorff’s α as the tag-level agreement estimate. Figure 3 presents the results of this metric, treating each LLM as a fourth coder and recalculating the agreement α . The figure plots one point for each model (Claude and GPT), representing the maximum α achieved across all shot-by-temperature combinations. Grey bars indicate human inter-rater agreement among the three original coders; the dashed vertical line marks $\alpha = 0.80$, which is the standard threshold in the literature for agreement among classifiers.

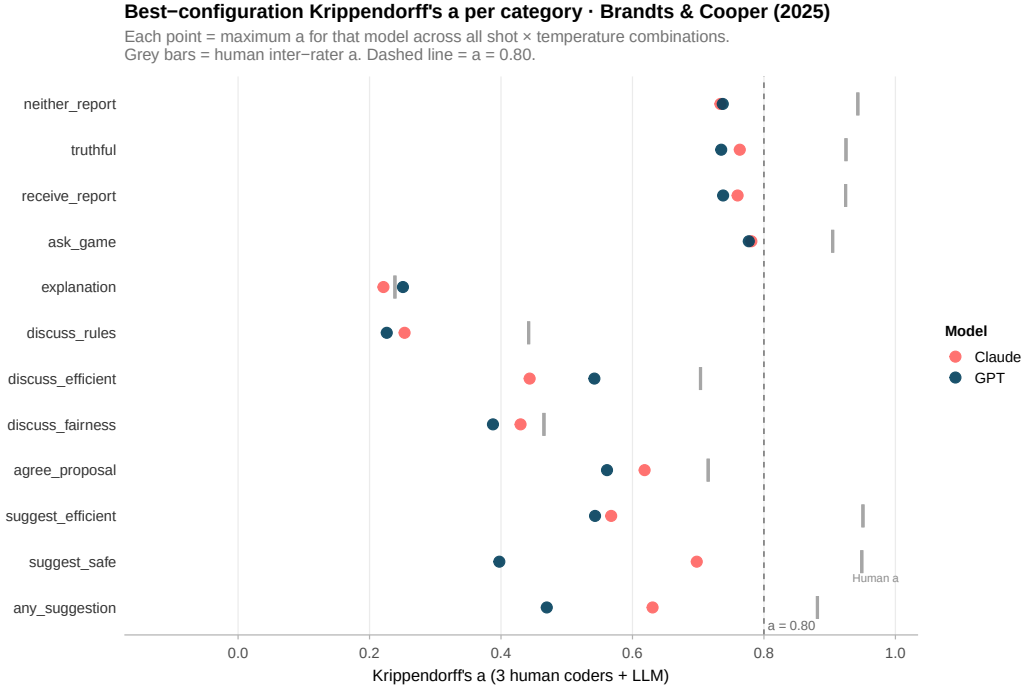


Figure 3. Best-configuration Krippendorff's α per category - Brandts and Cooper (2025)
Note: Each point = maximum α for that model across all shot $\times$ temperature combinations. Grey bars = human inter-rater α . Dashed line = $\alpha = 0.80$.

Among the 12 evaluable tags, agreement with human coders varies substantially across models. Claude achieves the highest best-configuration α for seven tags. The two LLMs perform poorly on *discuss_rules* and *discuss_fairness*. Human agreement is also modest for these two tags, suggesting that these categories are inherently ambiguous rather than uniquely difficult for LLMs to classify. Figure 4 reports the average Krippendorff's α across the 12 evaluable tags for each model, with configurations defined by prompting strategy (zero-shot versus few-shot) and temperature ($T \in 0, 0.1, 0.5, 1, 1.2$). Both models achieve their strongest performance under few-shot prompting at $T = 0$, with Claude attaining the higher mean α across categories. Variation across temperature settings within a given prompting strategy is modest, suggesting that performance in this task is relatively insensitive to LLM's temperature.

In this paper, we find that LLMs do not perform particularly well as substitutes for human coders. A key challenge is the complexity of the underlying data, which consist of three-person chat transcripts containing substantial jargon and implicit meanings that are difficult to interpret. Even among the human coders, agreement was relatively low for several categories, suggesting that many of the coding decisions are inherently subjective. In other words, the categories examined in this paper are likely to be interpretively ambiguous. For example, categories such as *Suggest safe*, *Discuss fairness*, and *Agree proposal* require coders

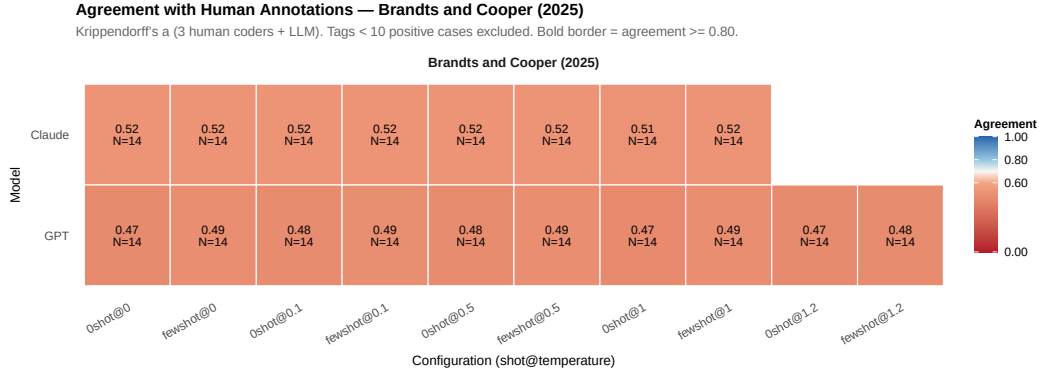


Figure 4. Overall agreement with Human Annotations - Brandts and Cooper (2025)
Note: Krippendorff's α (3 human coders + LLM). Tags < 10 positive cases excluded.

to infer speakers' intentions or pragmatic meaning rather than simply identify explicit textual features. Consequently, LLMs and human coders may be applying different implicit coding criteria, leading to systematic differences in their classifications.

To assess the consistency of response classifications across temperature settings, we treat each temperature run as an independent rater and compute Krippendorff's alpha over the resulting label vectors. This internal consistency analysis abstracts from agreement with the human benchmark and instead evaluates whether a model's coding decisions remain stable across temperature variations. High alpha values indicate that the model produces consistent classifications regardless of temperature, whereas low values suggest that temperature meaningfully affects classification outcomes for a given category. Table 1 reports label-level alpha values for Claude (temperatures 0, 0.1, 0.5, and 1) and GPT (temperatures 0, 0.1, 0.5, 1, and 1.2) under both zero-shot and few-shot prompting conditions.

Table 1. Internal consistency across temperature settings – Brandts and Cooper (2025)

Category	Claude		GPT	
	Zero-shot	Few-shot	Zero-shot	Few-shot
Any suggestion	0.920	0.944	0.849	0.866
Safe suggestion	0.656 ^b	0.877	0.378 ^b	0.401 ^b
Efficient suggestion	0.767 ^a	0.846	0.737 ^a	0.719 ^a
Agreement	0.915	0.940	0.813	0.824
Discuss fairness	0.938	0.891	0.783 ^a	0.602 ^b
Discuss efficiency	0.871	0.802	0.745 ^a	0.693 ^a
Discuss rules	0.787 ^a	0.789 ^a	0.388 ^b	0.499 ^b
Explanation	0.889	0.865	0.720 ^a	0.707 ^a
Discuss how to play	0.757 ^a	0.844	0.557 ^b	0.418 ^b
Ask about game	0.922	0.962	0.735 ^a	0.795 ^a
Receive report	0.916	0.933	0.736 ^a	0.753 ^a
Truthful about game	0.917	0.933	0.731 ^a	0.746 ^a
Lie about game	0.845	0.892	0.575 ^b	0.519 ^b
Contradiction	0.799 ^a	0.745 ^a	0.267 ^b	0.610 ^b
Neither option	0.902	0.960	0.736 ^a	0.754 ^a
<i>Mean</i>	<i>0.853</i>	<i>0.881</i>	<i>0.650</i>	<i>0.660</i>

Note: Krippendorff’s alpha treating each temperature run as an independent rater (predictions binarized: $>0.5 \rightarrow 1$, else 0). Claude: 4 temperature settings (0, 0.1, 0.5, 1); GPT: 5 settings (0, 0.1, 0.5, 1, 1.2). ^a Tentative agreement ($0.67 \leq \alpha < 0.80$). ^b Unreliable agreement ($\alpha < 0.67$). All unmarked values indicate strong agreement ($\alpha \geq 0.80$).

Table 1 shows that Claude achieves strong internal consistency across most categories, with mean $\alpha = 0.853$ under zero-shot prompting and $\alpha = 0.881$ under few-shot prompting. Under zero-shot prompting, only *safe suggestion* falls below the 0.67 reliability threshold, while four categories fall within the tentative agreement range. Few-shot prompting reduces the number of tentative categories to two: *discuss rules* and *contradiction*. Conversely, GPT exhibits substantially lower internal consistency (mean $\alpha = 0.650$ under zero-shot prompting and $\alpha = 0.660$ under few-shot prompting), with four categories falling below 0.67 in the zero-shot setting. Few-shot prompting does not consistently improve GPT’s stability. These results indicate that Claude’s classifications are largely insensitive to temperature variation, whereas GPT’s coding decisions, particularly for categories requiring pragmatic inference about conversational context, vary considerably across temperature settings.

5.1.2 Brandts and Cooper (2025): substantive validity

To evaluate substantive validity, that is, whether LLM-coded data lead to the same conclusions as human-coded data, we link both human- and LLM-generated labels to the

replication outcomes at the conversation-round level and restrict attention to the free-chat treatments. This analysis relies on files provided by the authors through their replication package and subsequent correspondence. The shared chat files do not include CH/A-D sessions, although the coding file contains CH/A-D observations for which no corresponding message-level source data are available. Because these source data are absent from the shared materials, the corresponding observations cannot be classified by the LLMs. Accordingly, all LLM-based analyses are restricted to the treatments covered by the shared files, and results requiring CH/A-D observations fall outside the scope of this analysis. For all analyses, we use the few-shot prompting configuration at temperature $T = 0$.

We then replicate the core regression analysis of [Brandts and Cooper \(2025\)](#), which examines how communication content predicts coordination outcomes. Table 8 in the original paper reports Probit regressions in which binary coordination outcomes, whether a group coordinated, and whether coordination was efficient, are regressed on the chat content categories, separately by treatment. We replicate this specification as a Probit model on the restricted shared-data sample, replacing the human-coded categories with LLM-generated categories in turn, so that all columns (human, Claude, GPT, ensemble) are estimated on the same observations.

Because we replicate the specifications of the original studies, we report unadjusted p -values throughout. Our objective is to assess whether a researcher substituting LLM-generated codes into the published analysis would reach the same substantive conclusions. Doing so requires holding the inference procedure fixed to match that of the original study; applying a multiplicity correction to the LLM-generated estimates but not to the human-coded benchmark would undermine the comparison we seek to make.

Table 2 reports the results for the CH/S-D coordination panel, while Tables 3 and 4 report the results for the CH/S-D efficient coordination and CH-MC efficient coordination panels, respectively. In the *CH/S-D coordination* panel, the human-coded benchmark indicates a positive and statistically significant effect of *Agreement* ($\beta = 0.196$, $p = .002$), broadly consistent with the original benchmark estimate ($\beta = 0.108$). GPT-based coding also yields a positive coefficient on *Agreement* ($\beta = 0.057$, $p = .103$), although the estimate is not statistically significant at the 5% level. By contrast, Claude-based coding identifies a positive and statistically significant effect of *Questions about play* ($\beta = 0.050$, $p < .001$), a relationship that is absent from the human-coded benchmark.

In the *CH/S-D efficient coordination* panel, the human-coded benchmark remains broadly consistent with the pattern reported in the original paper. Claude-based coding yields positive and statistically significant effects for *Discuss need to coordinate* ($\beta = 0.290$, $p = .007$), *Discuss fairness* ($\beta = 0.138$, $p = .002$), and *Discuss efficiency* ($\beta = 0.171$, $p = .047$),

whereas GPT-based coding does not produce statistically significant effects in this specification.

Finally, in the *CH-MC efficient coordination* panel, the human-coded benchmark again closely matches the replication-package estimate for *Truthfully reveal game* ($\beta = 0.290$, $p < .001$; paper estimate: 0.295). GPT-based coding produces a negative and statistically significant effect for *Explanation* ($\beta = -0.171$, $p = .037$) and a positive and statistically significant effect for *Lie about game* ($\beta = 0.432$, $p < .001$). The ensemble specification also yields a positive and statistically significant effect for *Lie about game* ($\beta = 0.261$, $p = .015$).

Overall, the Probit regressions lead to the same qualitative conclusion: LLM-derived codings do not consistently reproduce the human-coded benchmark coefficients in this application. Some alignment emerges for selected variables, for example, the positive association between *Agreement* and CH/S-D coordination in the GPT-based specification, but several statistically significant LLM-derived coefficients differ from the human benchmark in either sign or substantive prominence. This pattern is consistent with the agreement analysis discussed above: categories characterized by greater interpretive ambiguity tend to produce less stable downstream substantive inferences.

Table 2. Probit marginal effects on the restricted shared-data sample: CH/S-D, coordination.

Dependent variable	CH/S-D, coordination			
	Human	Claude	GPT	Ensemble
Agreement	0.196** [0.063]	0.028 [0.036]	0.057 [0.035]	0.058 [0.051]
Discuss need to coordinate	-0.047 [0.081]	-0.023 [0.046]	0.015 [0.026]	0.026 [0.022]
Discuss fairness	0.010 [0.021]	0.003 [0.014]	-0.027 [0.021]	0.005 [0.019]
Discuss efficiency	0.010 [0.026]	0.011 [0.014]	-0.017 [0.034]	-0.027 [0.041]
Questions about rules	-0.085 [0.067]	-0.228 [0.163]	-0.012 [0.065]	-0.042 [0.106]
Questions about play	0.025 [0.021]	0.050*** [0.015]	-0.310 [0.194]	-0.188 [0.193]
Explanation	-0.053 [0.043]	0.011 [0.021]	0.026 [0.030]	0.027 [0.033]
Observations	458	458	458	458
Pseudo- R^2	0.276	0.168	0.195	0.184

Notes: Entries are Probit marginal effects (evaluated at means) with clustered standard errors at the conversation-group level, reported in brackets. Pseudo- R^2 is deviance-based. Significance: * $p < .05$, ** $p < .01$, *** $p < .001$, p -values are unadjusted, following the specification of the original study.

All columns are estimated on the same restricted shared-data sample; the human column is the apples-to-apples benchmark. LLM estimates use few-shot prompting at temperature 0 (Claude, GPT). CH/A-D is omitted because message-level inputs are not present in the shared classification files. Blank cells indicate regressors dropped due to no within-sample variation.

Table 3. Probit marginal effects on the restricted shared-data sample: CH/S-D, efficient coordination.

Dependent variable	CH/S-D, efficient coordination			
	Human	Claude	GPT	Ensemble
Agreement	0.201*** [0.053]	0.031 [0.083]	0.056 [0.056]	−0.009 [0.084]
Discuss need to coordinate	0.206 [0.166]	0.290** [0.108]	−0.039 [0.106]	−0.010 [0.096]
Discuss fairness	−0.252*** [0.075]	0.138** [0.044]	0.100 [0.066]	0.142** [0.053]
Discuss efficiency	0.252*** [0.065]	0.171* [0.086]	−0.007 [0.067]	0.063 [0.072]
Questions about rules	−0.060 [0.150]	0.103 [0.210]	−0.012 [0.229]	0.155 [0.152]
Questions about play	−0.106 [0.139]	−0.180 [0.189]	0.064 [0.216]	−0.104 [0.212]
Explanation	0.056 [0.125]	−0.125 [0.071]	0.032 [0.064]	−0.062 [0.059]
Observations	458	458	458	458
Pseudo- R^2	0.208	0.181	0.163	0.168

Notes: Entries are Probit marginal effects (evaluated at means) with clustered standard errors at the conversation-group level, reported in brackets. Pseudo- R^2 is deviance-based. Significance: * $p < .05$, ** $p < .01$, *** $p < .001$, p -values are unadjusted, following the specification of the original study.

All columns are estimated on the same restricted shared-data sample; the human column is the apples-to-apples benchmark. LLM estimates use few-shot prompting at temperature 0 (Claude, GPT). CH/A-D is omitted because message-level inputs are not present in the shared classification files. Blank cells indicate regressors dropped due to no within-sample variation.

Table 4. Probit marginal effects on the restricted shared-data sample: CH-MC, efficient coordination.

Dependent variable	CH-MC, efficient coordination			
	Human	Claude	GPT	Ensemble
Agreement	0.044 [0.083]	0.029 [0.087]	−0.008 [0.073]	0.016 [0.095]
Discuss need to coordinate	−0.090 [0.190]	−0.059 [0.206]	−0.119 [0.158]	−0.146 [0.140]
Discuss fairness	−0.101 [0.092]	0.002 [0.055]	0.091 [0.063]	0.022 [0.057]
Discuss efficiency	0.109 [0.088]	0.066 [0.056]	0.151 [0.094]	0.149 [0.107]
Questions about rules	−0.166 [0.110]	−0.099 [0.114]	0.045 [0.142]	−0.033 [0.096]
Questions about play	−0.118 [0.157]	0.056 [0.147]	0.156 [0.106]	0.063 [0.097]
Explanation	0.034 [0.100]	−0.012 [0.062]	−0.171* [0.082]	−0.082 [0.094]
Ask what game	0.118 [0.083]	0.050 [0.094]	0.046 [0.070]	0.016 [0.080]
Truthfully reveal game	0.290*** [0.081]	0.028 [0.080]	−0.063 [0.051]	0.043 [0.072]
Lie about game	−0.032 [0.198]	0.191 [0.148]	0.432*** [0.045]	0.261* [0.107]
Observations	458	420	420	420
Pseudo- R^2	0.283	0.251	0.281	0.261

Notes: Entries are Probit marginal effects (evaluated at means) with clustered standard errors at the conversation-group level, reported in brackets. Pseudo- R^2 is deviance-based. Significance: * $p < .05$, ** $p < .01$, *** $p < .001$, p -values are unadjusted, following the specification of the original study.

All columns are estimated on the same restricted shared-data sample; the human column is the apples-to-apples benchmark. LLM estimates use few-shot prompting at temperature 0 (Claude, GPT). CH/A-D is omitted because message-level inputs are not present in the shared classification files. Blank cells indicate regressors dropped due to no within-sample variation.

5.2 Application 2: Messages and promises in a trust game over time (Ederer and Schneider, 2022)

The second application we consider is Ederer and Schneider (2022). The authors examine how temporal delay affects trust, trustworthiness, and the credibility of promises, motivated by the empirical observation that trust often involves delayed reciprocation. Using a large-scale hybrid laboratory–online experiment with over 700 participants, they employ an adaptation of the trust game (Berg et al., 1995; Güth et al., 1997), similar to that proposed by Charness and Dufwenberg (2006), by introducing exogenous time delays between the trustor’s decision to send money and the trustee’s opportunity to reciprocate. The experimental design varies three temporal conditions: immediate decisions during the laboratory session, early decisions within 24 hours, and late decisions after a 21-day delay. Crucially for

our purposes, they incorporate a pre-play communication stage that allows trustees to send free-form messages, including non-binding promises, to trustors. This design allows for an examination of whether the positive effects of communication on cooperative behavior persist when reciprocation is delayed, testing the hypothesis that temporal distance may erode both the propensity to honor commitments and the overall level of trustworthiness.

The experiment measures trust by the amount sent by trustors and trustworthiness by the amount returned by trustees, with trustees making 10 binary ROLL/DON'T ROLL choices, one of which is randomly implemented. To analyze free-form communication, the authors employ two hypothesis-blind research assistants to independently classify messages according to whether they contain an explicit promise to choose ROLL (e.g., "I will choose ROLL" or "I promise to choose ROLL").

5.2.1 Ederer and Schneider (2022): validation metrics

In the first application, we used Claude and GPT as the evaluated LLMs. In the subsequent two applications, we additionally included Gemini. Gemini is substantially more expensive, costing approximately two to three times more than Claude and up to five times more than GPT. Because the first application involved a more computationally demanding dataset, we restricted the use of Gemini to the second and third applications.

For this paper, we do not report Krippendorff's alpha because only one human classification is available for each tag. Accordingly, we focus on Cohen's kappa as the primary agreement metric. Figure 5 presents the Cohen's kappa values for the three analyzed categories. As in Application 1, we select the best-performing configuration for each model across all combinations of prompting strategy (zero-shot versus few-shot) and temperature, and report the resulting Cohen's kappa as the tag-level agreement estimate. The figure displays one point for each model (Claude, Gemini, and GPT), corresponding to the maximum κ achieved across all shot-by-temperature combinations. The dashed vertical line marks $\kappa = 0.80$.

Figure 5 shows that the categories `is_promise` and `is_efficiency` achieve high agreement between LLMs and human coders, with the exception of Gemini on `is_promise`. In contrast, the category `is_ethics` exhibits lower Cohen's kappa values across models.

Overall, the categories in this application correspond to relatively well-defined normative concepts that language models can identify from the surface and semantic features of the text. Unlike Application 1, classification in this setting does not require tracking conversational context or inferring pragmatic intent. Moreover, the dataset does not suffer from severe class imbalance, meaning that each category contains a sufficient number of

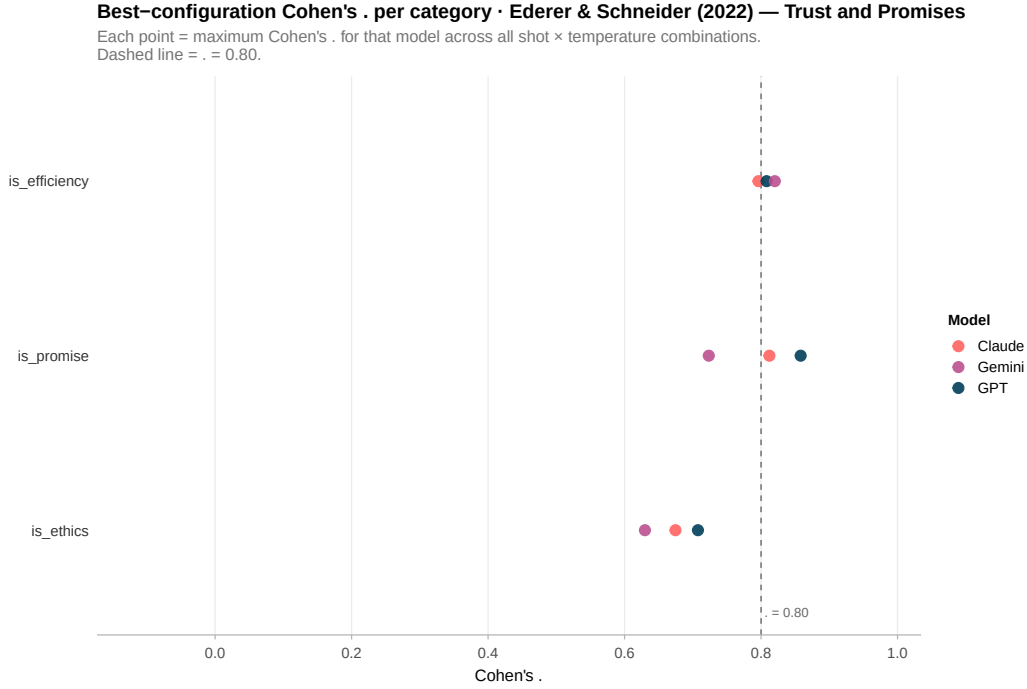


Figure 5. Best-configuration Cohen's κ per category - Ederer and Schneider (2022)
Note: Each point = maximum κ for that model across all shot $\times$ temperature combinations. Grey bars = human inter-rater κ . Dashed line = $\kappa = 0.80$.

positive cases to support meaningful agreement estimates.

The comparatively lower agreement for *is_ethics* may indicate that LLMs tend to over-predict ethical content, labeling statements as ethical when the human coder did not. As a result, the mean agreement values reported in Figure 6 fall below the 0.80 threshold, with the lower performance of this category reducing overall agreement levels. This asymmetry is noteworthy because ethics is likely the most interpretation-dependent of the three categories.

Finally, Table 5 reports internal consistency across temperature settings for the trust-promises application. All model-strategy combinations achieve strong Krippendorff's alpha values, with no entry falling below the 0.80 threshold. Claude is the most consistent model, attaining near-perfect agreement under zero-shot prompting (mean $\alpha = 0.994$), with *promise* and *ethics* reaching 1.000 and *efficiency* reaching 0.982. Few-shot prompting yields similarly high stability (mean $\alpha = 0.979$). Gemini is likewise highly consistent, with mean alpha values of 0.902 under zero-shot prompting and 0.918 under few-shot prompting, although the *ethics* category is somewhat less stable (0.845 under zero-shot prompting and 0.870 under few-shot prompting). We can also see in Figure 6 that Gemini was the worse performing LLM.

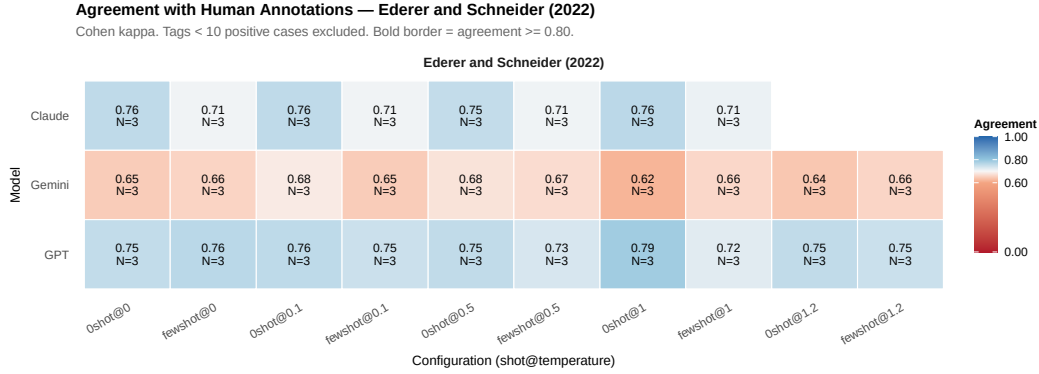


Figure 6. Overall agreement with Human Annotations - Ederer and Schneider (2022)
Note: Cohen’s κ (human coder + LLM).

The near-ceiling consistency observed across all model-strategy combinations likely reflects the conceptual clarity of the three coding categories. A message either contains an explicit promise, an efficiency argument, or an ethical argument, each of which can be identified from surface-level semantic features of the text without requiring substantial inference about conversational context or pragmatic intent. Overall, this application provides the clearest example of the conditions under which LLM-based coding performs well: a small number of well-defined, non-overlapping categories with balanced class representation.

Table 5. Internal consistency across temperature settings – Ederer and Schneider (2022)

Category	Claude		GPT		Gemini	
	Zero-shot	Few-shot	Zero-shot	Few-shot	Zero-shot	Few-shot
Promise	1.000	0.990	0.978	0.969	0.900	0.980
Efficiency	0.982	0.983	0.984	0.957	0.962	0.902
Ethics	1.000	0.965	0.900	0.904	0.845	0.870
<i>Mean</i>	<i>0.994</i>	<i>0.979</i>	<i>0.954</i>	<i>0.943</i>	<i>0.902</i>	<i>0.918</i>

Note: Krippendorff’s alpha treating each temperature run as an independent rater (predictions binarized: $>0.5 \rightarrow 1$, else 0). Claude: 4 temperature settings (0, 0.1, 0.5, 1). GPT and Gemini: 5 settings (0, 0.1, 0.5, 1, 1.2). All values indicate strong agreement ($\alpha \geq 0.80$).

5.2.2 Ederer and Schneider (2022): substantive validity

To assess whether the same substantive conclusions emerge when using LLM-coded data instead of human-coded data, we use the replication package provided by the authors on the publication website. Among the three coded labels, only *is_promise* is used in the published analysis of Ederer and Schneider (2022). The authors code whether each trustee message contains an explicit promise to choose ROLL and use this classification in a single core regression: an OLS specification of *numrolls* (the number of ROLL choices out of 10)

on a communication-treatment indicator and *is_promise*, restricted to trustees (*role* = 1, $N = 354$), with standard errors clustered at the session level. In the original paper, this specification yields a positive and statistically significant effect for *is_promise*, indicating that trustees who explicitly promise to cooperate choose ROLL substantially more often than those who do not.

To address this question, we replace the human-coded *is_promise* variable with LLM-generated predictions from each model using the zero-shot, temperature-0 configuration, which achieved the strongest agreement performance for this label in the analysis above. For trustees in the no-communication condition, *is_promise* is set equal to 0 for all models, consistent with the original coding procedure, as these participants did not send a message and therefore could not make a promise. For the 172 trustees who sent messages, LLM predictions are merged using the message text as the matching key. The regression specification is otherwise identical to that used in the original paper.

Table 6 reports the results. The human-coded baseline closely reproduces the original finding ($\hat{\beta} = 1.977$, $p = .037$). These results may differ slightly from those reported in the original paper because some observations are lost during the classification process; accordingly, we restrict the analysis to the subset of observations also coded by the LLMs. Under LLM-based coding, the coefficient on *Promise* retains the same sign and a broadly similar magnitude across all three models, but it is not statistically significant at the 5% level in any of them. The point estimate is closest to the original for GPT ($\hat{\beta} = 1.746$, $p = .060$), followed by Gemini ($p = .085$) and Claude ($p = .128$). The communication-treatment indicator is not statistically significant in any specification, consistent with the original paper.

Table 6. Regression comparison: human vs. LLM-coded if the trustee made a promise – Ederer and Schneider (2022)

	Human	Claude	GPT	Gemini
Communication	0.521 [0.813]	0.919 [0.762]	0.723 [0.733]	0.709 [0.787]
Promise	1.977* [0.946]	1.454 [0.957]	1.746 [0.928]	1.507 [0.875]
Constant	3.652*** [0.319]	3.652*** [0.319]	3.652*** [0.319]	3.652*** [0.319]
N	353	353	353	353
R^2	0.065	0.056	0.061	0.054

Note: OLS with standard errors clustered by session. Sample: trustees (*role* = 1). Human column uses original author coding. LLM columns replace *is_promise* with model predictions at zero-shot, temperature 0. Standard errors in brackets. * $p < .05$, ** $p < .01$, *** $p < .001$.

Overall, these results suggest that when the classification task is relatively straightforward, LLM-based coding can recover the same qualitative substantive conclusions as human coding. Although the LLM-based estimates exhibit greater noise, leading in some cases to weaker statistical significance, the direction and approximate magnitude of the effects remain broadly consistent with the human-coded benchmark.

The remaining two labels, *is_efficiency* and *is_ethics*, appear only in the appendices as descriptive statistics. Efficiency appeals are present in 49 per cent of messages and are strongly correlated with promise-making ($r = 0.75$), whereas ethics appeals appear in only 8 per cent of messages.

Table 7 extends the comparison to the secondary labels reported descriptively in Appendix D of Ederer and Schneider (2022). Panel A reports prevalence rates among the 172 trustees in the communication condition. For *Promise*, Claude and GPT produce estimates close to the human benchmark (67.4% versus 69.8%), whereas Gemini overpredicts the category (79.1%). For *Efficiency*, all three models substantially overpredict relative to the human coder (57-59% versus 49%), suggesting that LLMs identify efficiency-related language more broadly than the comparatively conservative human coding standard. The divergence is largest for *Ethics*: the human coder identifies ethics appeals in only 7.6% of messages, compared with 12.2% for Claude, 9.9% for GPT, and 22.1% for Gemini. This pattern is consistent with a tendency for LLMs to classify implicit normative language as ethical framing even when a human coder would not interpret it as an explicit ethics appeal.

Panel B reports Pearson correlations for the full trustee sample ($N = 354$). The correlation between *Promise* and *Efficiency* is closely reproduced by Claude ($r = 0.699$ versus the human benchmark of 0.709) and reasonably approximated by GPT ($r = 0.671$), while Gemini produces a somewhat higher estimate ($r = 0.774$). By contrast, the correlation between *Promise* and *Ethics* is more distorted. The human benchmark of $r = 0.146$ increases to 0.181 under Claude, 0.237 under GPT, and 0.383 under Gemini, reflecting the tendency of the models, particularly Gemini, to overpredict ethics-related labels and thereby inflate apparent co-occurrence.

Table 7. Label prevalence and correlations: human vs. LLM coding – [Ederer and Schneider \(2022\)](#)

	Human	Claude	GPT	Gemini
<i>Panel A: Prevalence (% of 172 communication-condition messages)</i>				
Promise	69.8	67.4	67.4	79.1
Efficiency	48.8	59.3	57.0	58.7
Ethics	7.6	12.2	9.9	22.1
<i>Panel B: Pearson correlations (full trustee sample, $N = 354$)</i>				
$r(\text{Promise, Efficiency})$	0.709	0.699	0.671	0.774
$r(\text{Promise, Ethics})$	0.146	0.181	0.237	0.383

Note: LLM predictions at zero-shot, temperature 0. Prevalence computed among the 172 communication-condition trustees; 13 rows with missing messages are assigned label 0 (matching human coding). Correlations computed on the full trustee sample ($N = 354$, including no-communication trustees with all labels set to 0).

5.3 Application 3: Social Desirability Bias and AI use ([Ling et al., 2025](#))

The third application draws on data from [Ling et al. \(2025\)](#), in which the authors study the discrepancy between self-reported AI usage and perceived peer usage in educational settings. The authors attribute this gap primarily to social desirability bias ([Krumpal, 2013](#)). The study employs an indirect questioning method, commonly used in the psychology literature ([Fisher, 1993](#)), with 338 university students, by asking respondents to report both their own AI use and their perceptions of peers’ usage.

To investigate the mechanisms underlying this discrepancy, the researchers collected open-ended responses from a separate follow-up survey involving 96 undergraduate students recruited via Prolific. This sample serves as the benchmark for our analysis. Participants were presented with findings from the initial survey, highlighting the reporting gap, and asked an open-ended question to explain it in their own words. These responses were analyzed using a streamlined thematic analysis approach: the first author conducted open coding to identify proposed explanations, after which all authors jointly refined and agreed upon a set of inductive codes, which were then iteratively applied to classify each response, with ambiguous cases resolved through author consultation. This coding process identifies social desirability bias as the most frequently cited explanation for the reporting gap, while also surfacing alternative mechanisms such as overestimation of peers’ AI usage driven by availability bias and information asymmetry regarding actual adoption rates among peers. This application is particularly well-suited for evaluating whether LLMs can reproduce inductive thematic coding on short explanatory responses.

5.3.1 Ling et al. (2025): validation metrics

For this application we also use Cohen's kappa as the primary agreement metric. The coding categories used by Ling et al. (2025) capture twelve distinct mechanisms that may explain the gap between personal and peer-reported AI usage. *Uninformative* responses contain nonsensical or contradictory content that misunderstands the question. *SDB* (Social Desirability Bias) refers to the possibility that individuals misreport or conceal AI usage to avoid negative social evaluation, judgment, or reputational harm. *Academic_integrity* captures concerns that AI usage constitutes cheating, dishonesty, or academically inappropriate behavior. *Overest_others* indicates that students overestimate their peers' AI usage, whereas *Underest_own* reflects students underestimating their own usage. *Info_asymmetry* captures students' limited knowledge about their peers' actual AI usage patterns. *Privacy_concerns* reflects reluctance to disclose personal AI usage because of privacy considerations. *Self_esteem* indicates that students underreport usage to protect their internal self-image, distinct from concerns about external judgment captured by *SDB*. *Self_report_bias* captures broader self-enhancement or self-favoring biases in reported behavior. *AI_discussion_priming* attributes the gap to confusion between the visibility and frequency of AI-related discussions or media coverage and the actual prevalence of AI usage. *Network_effect* explains the gap by suggesting that students generalize from their local social networks, inferring broader population-level AI usage from the behavior observed within their immediate peer group. Finally, *Truthful* indicates that the reported gap reflects genuine differences between personal and peer AI usage.

Figure 7 presents the Cohen's kappa values for the twelve coding categories. As in the previous applications, we select the best-performing configuration for each model across all combinations of prompting strategy (zero-shot versus few-shot) and temperature, and report the resulting Cohen's kappa as the tag-level agreement estimate. The figure displays one point for each model (Claude, Gemini, and GPT), corresponding to the maximum κ achieved across all shot-by-temperature combinations. The dashed vertical line marks $\kappa = 0.80$, the conventional threshold for strong agreement.

Figure 7 shows substantial heterogeneity in agreement across categories. Four categories, *overestimate peers*, *social desirability bias*, *academic integrity*, and *information asymmetry*, reach or closely approach the $\kappa = 0.80$ threshold for strong agreement between the human coding and each LLM. This pattern is consistent with the explicit theoretical grounding of these constructs and the relative clarity with which they are expressed in respondents' answers.

In contrast, *self-report bias*, *network effect*, and *privacy concerns* remain well below the threshold for every model, even under the best-performing configuration. This likely reflects

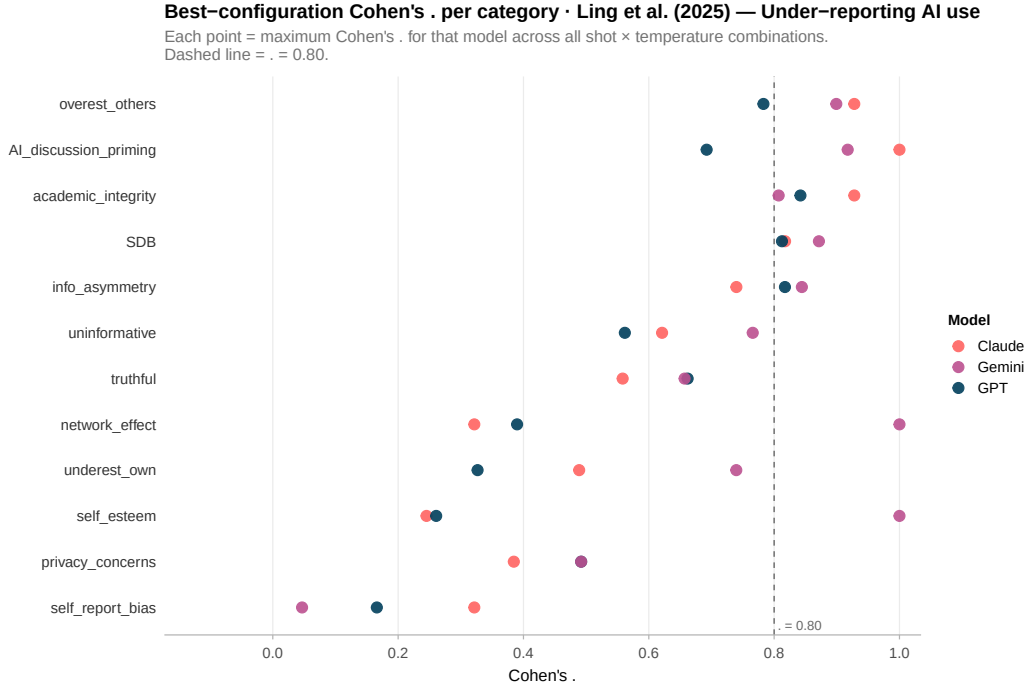


Figure 7. Best-configuration Cohen's κ per category - Ling et al. (2025)

Note: Each point = maximum κ for that model across all shot $\times$ temperature combinations. Grey bars = human inter-rater κ . Dashed line = $\kappa = 0.80$.

both the more inductive origins of these labels and the inferential demands they place on coders, who must attribute implicit motivations to short, open-ended responses.

A notable outlier is the *self-esteem* category, for which Gemini achieves near-perfect agreement ($\kappa \approx 1.0$), whereas Claude and GPT both fall below 0.25. Such extreme divergence is more plausibly attributable to Gemini collapsing class variation by systematically predicting a single value than to genuine alignment with human judgment. Class-balance diagnostics should therefore be examined before interpreting this result as evidence of superior performance.

Figure 8 presents Cohen's κ relative to the human benchmark across all ten shot $\times$ temperature configurations for the five categories meeting the minimum prevalence threshold. Gemini achieves the highest and most stable agreement, with κ values ranging narrowly from 0.73 to 0.78 and showing minimal sensitivity to prompting strategy or temperature. Claude performs similarly under zero-shot prompting ($\kappa = 0.70$ – 0.73) but deteriorates under few-shot prompting at lower temperatures, suggesting that the examples may introduce an interfering prior. GPT exhibits the greatest variability, with κ ranging from 0.39 to 0.74, although few-shot prompting generally improves performance relative to zero-shot. Overall, even the best-performing configurations achieve only moderate agreement, reflecting the inductive nature of the coding scheme for some categories, which requires inference about

social motivations rather than identification of surface-level linguistic features.

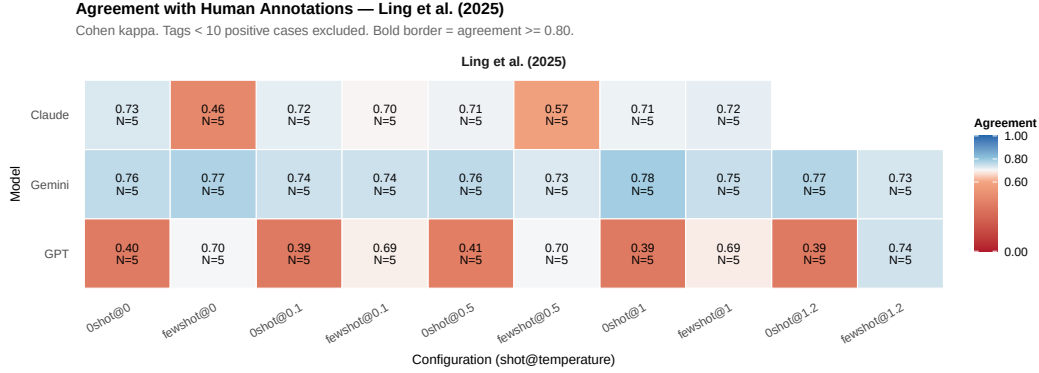


Figure 8. Overall agreement with Human Annotations - Ling et al. (2025)
Note: Cohen’s κ (human coder + LLM).

Table 8 reports internal consistency across temperature settings for the under-reporting application. All three models achieve strong mean alpha values, although notable variation emerges for specific labels. Claude attains the highest and most uniform consistency, with mean $\alpha = 0.950$ under zero-shot prompting and $\alpha = 0.973$ under few-shot prompting; three labels exhibit no cross-temperature variance in a given strategy (shown as —) and are therefore excluded from the mean calculation. Gemini is also broadly consistent (mean $\alpha = 0.825$ under zero-shot prompting and $\alpha = 0.827$ under few-shot prompting), although two labels fall below the reliability threshold: *underestimate self* ($\alpha = 0.621$ under zero-shot prompting) and *network effect* ($\alpha = 0.527$ under zero-shot prompting; 0.604 under few-shot prompting). GPT achieves strong mean consistency under zero-shot prompting ($\alpha = 0.885$), with only *self-report bias* (0.752) and *truthful response* (0.768) falling within the tentative agreement range and no category below 0.67 . Notably, the problematic labels for Gemini (*underestimate self* and *network effect*) are among the most inductively defined categories in the coding scheme, suggesting that nuanced social-psychological constructs are more sensitive to temperature variation than more concrete thematic labels such as *social desirability bias* or *academic integrity*.

Table 8. Internal consistency across temperature settings – Ling et al. (2025)

Category	Claude		Gemini		GPT
	Zero-shot	Few-shot	Zero-shot	Few-shot	Zero-shot
Uninformative	0.974	1.000	0.859	0.920	0.927
Social desirability bias	0.958	1.000	0.966	0.954	0.979
Overestimate peers	0.950	1.000	0.899	0.884	0.808
Underestimate self	0.882	0.884	0.621 ^b	0.825	0.798 ^a
Academic integrity	1.000	1.000	0.912	0.848	0.953
Information asymmetry	0.968	1.000	0.832	0.904	0.893
AI discussion priming	—	1.000	—	0.861	0.916
Privacy concerns	0.901	—	1.000	1.000	1.000
Self-esteem	0.936	1.000	0.889	0.803	0.893
Self-report bias	—	0.813	0.846	0.486 ^b	0.752 ^a
Network effect	0.963	1.000	0.527 ^b	0.604 ^b	0.935
Truthful response	0.969	1.000	0.725 ^a	0.834	0.768 ^a
<i>Mean</i> [†]	<i>0.950</i>	<i>0.973</i>	<i>0.825</i>	<i>0.827</i>	<i>0.885</i>

Note: Krippendorff’s alpha treating each temperature run as an independent rater (predictions binarized: $>0.5 \rightarrow 1$, else 0). Claude: 4 temperature settings (0, 0.1, 0.5, 1); Gemini and GPT: 5 settings (0, 0.1, 0.5, 1, 1.2). GPT results are zero-shot only for this application. ^a Tentative agreement ($0.67 \leq \alpha < 0.80$). ^b Unreliable agreement ($\alpha < 0.67$). Unmarked values: strong agreement ($\alpha \geq 0.80$). — no cross-temperature variance or insufficient data. [†] Mean excludes cells marked —.

5.3.2 Ling et al. (2025): substantive validity

Finally, as in the previous applications, we examine whether using LLM-coded data instead of the authors’ original classifications would lead to the same substantive conclusions. Ling et al. (2025) report category prevalence estimates showing that social desirability bias (*SDB*) is the dominant explanation for the own-other AI usage gap, appearing in 60.6% of valid responses, followed by overestimation of others’ usage (29.6%), information asymmetry (26.8%), and academic integrity concerns (21.1%). The key substantive validity question in this application is therefore whether LLM-based coding reproduces the rank ordering and relative magnitudes of these categories closely enough to preserve the paper’s qualitative conclusions.

Table 9 reports prevalence rates from the human coding and from each model under zero-shot prompting at temperature 0. The main conclusion of Ling et al. (2025), namely, that social desirability bias (*SDB*) is the dominant mechanism underlying the own-other AI usage gap, is robustly preserved across all three models. LLM-estimated *SDB* prevalence ranges from 57.5% (Claude) to 63.2% (Gemini), closely bracketing the human benchmark of 60.6%. Information asymmetry and academic integrity are likewise reproduced with

high fidelity, with no model deviating by more than five percentage points from the human benchmark for either category. Overall, these results suggest that the paper’s central substantive conclusions: that SDB is the primary driver of underreporting, that overestimation of peers’ AI usage plays a secondary role, and that information asymmetry represents a distinct explanatory mechanism, are reliably recovered through LLM-based coding.

Table 9. Category prevalence: human vs. LLM coding – [Ling et al. \(2025\)](#)

	Human	Claude	Gemini	GPT
<i>Panel A: High-prevalence categories (human $\geq 20\%$)</i>				
Social desirability bias	60.6	57.5	63.2	62.8
Overestimate others	29.6	22.5 ^a	23.7 ^a	24.4 ^a
Information asymmetry	26.8	31.2	27.6	25.6
Academic integrity	21.1	16.2	21.1	19.2
<i>Panel B: Moderate-prevalence categories (human 5–20%)</i>				
AI discussion priming	11.3	6.2 ^a	7.9 ^a	5.1 ^a
Underestimate own use	7.0	12.5 ^b	21.1 ^b	21.8 ^b
<i>Panel C: Rare categories (human $< 5\%$)</i>				
Privacy concerns	4.2	3.8	1.3	1.3
Self-esteem	4.2	25.0 ^b	14.5 ^b	24.4 ^b
Truthful response	2.8	11.2 ^b	7.9 ^b	3.8
Self-report bias	1.4	1.2	19.7 ^b	14.1 ^b
Network effect	1.4	10.0 ^b	1.3	9.0 ^b

Note: Prevalence reported as percentage of valid responses (human: $n = 71$; LLM: $n = 68$ –80 depending on model’s uninformative classification). LLM predictions at zero-shot, temperature 0. Rows ordered by human prevalence within each panel. ^a LLM under-detects relative to human benchmark (difference > 5 pp). ^b LLM over-detects relative to human benchmark (difference > 5 pp).

6 Discussion and conclusion

This paper provides evidence on whether large language models can reliably code the kinds of open-ended text data that arise in experimental economics research. The three applications reveal a consistent structural regularity: LLM coding reliability is not determined by model choice or temperature, but by the fit between the coding scheme and what LLMs can reliably infer. LLMs perform well when categories are anchored in explicit language contained in the text. They struggle when coding requires interpreting intent, conversational context, or themes that emerge from reading across responses, particularly when the models must analyze conversations containing substantial jargon. This pattern holds across models and prompting strategies, suggesting it reflects a property of the measurement problem

rather than a limitation of any particular model. The trust game of [Ederer and Schneider \(2022\)](#) illustrates the favorable case: three categories with clear semantic anchors, short messages, and no conversational context to track. The coordination chat of [Brandts and Cooper \(2025\)](#) illustrates the unfavorable case: multi-label coding of three-party conversations where intent must be inferred rather than directly identified from the text. The thematic coding of [Ling et al. \(2025\)](#) sits in between, with a dominant category that LLMs recover reliably and rare inductively derived categories that they systematically over-identify.

When a category appears in fewer than roughly 5-7% of responses, LLM-based prevalence estimates tend to be substantially upward biased, often by a factor of three to five. This occurs because LLMs classify text as belonging to a category based on semantic similarity, even when a human coder strictly applying the coding protocol would not assign the label. The problem is easy to miss: for high-prevalence categories, over-identification has little effect on estimates, but for low-prevalence categories it can substantially inflate measured prevalence.

Notably, neither model choice nor temperature is a first-order determinant of performance differences. Across the three applications, the three model families (Claude, Gemini, GPT) perform similarly under their best-performing configurations: average rank differences remain below 0.25, and no model dominates consistently across all categories. These findings carry a practical implication: researchers may gain relatively little from extensive model comparisons or hyperparameter tuning. The bottleneck lies in coding scheme design and prompt design rather than in LLM selection.

Beyond the question of what drives agreement, there is a prior question about what agreement actually means for research use. A central methodological contribution of this paper is the distinction between classification agreement and substantive validity. High agreement between an LLM and a human coder is necessary but not sufficient for the LLM-coded variable to be usable in downstream inference. Conversely, imperfect agreement does not automatically disqualify a variable for research use.

The trust game application of [Ederer and Schneider \(2022\)](#) illustrates the first failure mode. GPT achieves Cohen's $\kappa \approx 0.84$ on the promise label, commonly interpreted as strong agreement, yet this does not fully preserve the statistical significance of the estimated promise effect in the regression ($p = .060$ under GPT coding versus $p = .037$ in our human-coded replication). The attenuation is consistent with classical measurement error: even a modest rate of misclassification dilutes the estimated effect when mislabeled observations differ systematically in outcomes from correctly labeled observations. Researchers who rely solely on agreement statistics to evaluate LLM coding may underestimate this cost.

The converse failure mode appears in the multi-label conversational coding application

of Brandts and Cooper (2025). Several LLM-derived variables achieve moderate values of Krippendorff’s α in the agreement analysis, yet produce regression coefficients that are inconsistent with the human-coded benchmark in both sign and significance: some coefficients reverse sign, while others achieve significance for variables that are not significant under human coding.

A further implication follows from the attenuation pattern documented above. When LLM miscoding is approximately random rather than systematically correlated with the outcome, it introduces classical measurement error into the coded variable, biasing estimated coefficients toward zero and reducing statistical power relative to human coding. This generates an important asymmetry for statistical inference: a statistically significant result obtained under LLM coding can be interpreted conservatively. If an effect remains statistically detectable despite the additional noise introduced by imperfect classification, it would almost certainly remain detectable under cleaner human-coded measurement. LLM coding therefore provides a conservative lower bound on effect detectability in settings where the coding task is well-defined and miscoding is not directional. The converse does not hold: a null result under LLM coding is uninformative, since attenuation may have suppressed a real effect. However, for researchers seeking to establish that an association exists, a statistically significant result under LLM coding provides meaningful evidential weight. This property is most clearly illustrated in Application 2: despite Cohen’s κ values below 1.0 and attenuated significance, all three models recover the positive sign of the promise effect, and a researcher relying on any of them would reach the same qualitative conclusion as the original study.

These failure modes point toward a set of practical recommendations for researchers considering LLM-assisted coding. Because the largest determinant of performance is how well the prompt communicates the coding task, investment in prompt design yields greater returns than model comparisons or hyperparameter tuning: few-shot examples should emphasize ambiguous boundary cases that distinguish adjacent categories, not just clean prototypical instances. Any category whose human-coded prevalence falls below approximately 5% is at elevated risk of LLM over-detection and warrants targeted human review or adjudication before prevalence estimates are interpreted quantitatively. On the question of reproducibility, temperature 0 is the natural default: performance is largely insensitive to temperature across all three applications, and a deterministic setting eliminates stochastic analytical flexibility while allowing the coding procedure to be reproduced exactly from a shared prompt and model version. Researchers should select models based on accessibility and cost and redirect the saved effort toward prompt iteration and validation.

Several limitations qualify the scope of our conclusions. First, the results are sensitive to prompt specification. Although we iterated across multiple prompt versions for each

application and found no substantial performance differences, the space of feasible prompt specifications is effectively unbounded, and replications using different specifications may yield different results. Prompt design should therefore be treated as a methodological disclosure rather than an implementation detail: sharing the exact prompt used is a necessary condition for reproducibility. Second, all three applications involve text produced by university-educated participants in structured experimental settings involving WEIRD populations. Whether the performance patterns documented here extend to other languages, lower-literacy populations, or less constrained text environments remains an open question. Third, frontier LLMs evolve rapidly; our results characterize performance at a particular technological snapshot, and successor models may perform substantially differently. Our quantitative benchmarks should be interpreted accordingly rather than treated as stable benchmarks. Fourth, and most fundamentally, all three applications treat human-coded benchmarks as ground truth without independently verifying their accuracy. We adopt the convention standard in the inter-rater agreement literature, where trained human judgments serve as the reference against which LLM performance is measured. This assumption is practically necessary but not without consequences: research assistants make errors, apply coding rules inconsistently, and sometimes disagree with each other at rates comparable to the LLM-human disagreement we document. What our results establish is therefore not that LLMs code experimental text correctly in an absolute sense, but that they approximate human judgments sufficiently closely to be empirically useful.

More broadly, our results suggest that LLMs are best understood as a component in a human-machine coding workflow rather than autonomous replacements for human judgment. For many coding tasks in experimental economics, the most practical application is not full automation but LLM-assisted coding with targeted human review of the cases most likely to be misclassified: low-prevalence categories, pragmatically complex messages, and observations near category boundaries. This hybrid approach preserves the scalability and cost advantages of LLM coding while retaining human oversight in the settings where it is most needed. There is, however, a dimension in which LLM-based coding is strictly superior to purely human coding regardless of agreement levels: reproducibility. A coding procedure defined by a prompt and a model version can be shared, re-executed, and independently verified; one that relies on research assistants whose training, decisions, and disagreements go undocumented cannot. As the social sciences continue to grapple with replicability and analytical transparency, this audit trail may ultimately prove as consequential as the associated cost savings. The question for future work is not only whether LLMs can match human coders, but whether the combination of near-human agreement and full procedural transparency can raise the evidentiary standard for text-based measurement procedures in experimental research.

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